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Title: A seascape dichotomy in the role of small consumers for coral reef energy fluxes

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Appendix S1

TABLE S1: Metadata and summary for community collections performed around the island of Mo'orea, French Polynesia. SA = surface area.

Site	Zone	Depth	Year	SA	abundance	diversity	biomass	coral_cover	turf_cover
B1	B	7.1	2017	5.35	54	13	44.05	0.14	0.69
B10	B	3.1	2017	6.12	112	17	88.98	0.15	0.64
B11	B	2.1	2017	6.12	120	21	110.90	0.40	0.39
B12	F	15.6	2017	4.64	32	19	43.88	0.77	0.10
B13	F	14.4	2017	5.35	43	21	53.91	0.32	0.47
B14	B	1.7	2017	6.12	31	7	2.31	0.05	0.42
B15	F	9.8	2017	3.67	13	9	21.40	0.47	0.06
B16	B	3.3	2017	5.73	92	21	41.71	0.42	0.30
B17	F	16.6	2017	3.67	18	11	30.96	0.22	0.30
B18	B	2.8	2017	5.73	89	18	12.52	0.01	0.63
B19	F	16.6	2017	6.12	25	9	9.85	0.43	0.10
B2	F	13	2017	5.35	40	16	51.14	0.34	0.41
B20	B	2.9	2017	5.17	103	31	127.40	0.28	0.18
B21	F	17	2017	6.52	34	16	27.12	0.35	0.07
B22	F	12.4	2017	3.67	27	18	69.91	0.56	0.10
B23	F	13.5	2018	5.35	20	14	21.44	0.48	0.08
B24	F	16	2018	4.64	23	17	40.91	0.19	0.38
B25	F	14	2018	5.73	43	23	64.71	0.53	0.24
B26	F	13.5	2018	4.64	17	13	44.37	0.37	0.14
B27	F	13.5	2018	4.64	14	9	6.68	0.43	0.15
B28	B	2	2018	6.52	49	23	29.57	0.00	0.52
B29	B	2.1	2018	5.73	84	24	16.62	0.31	0.33
B3	F	12.9	2017	4.99	25	16	33.48	0.24	0.43
B30	B	1.9	2018	5.54	76	18	12.20	0.21	0.60
B31	B	2	2018	4.30	57	14	29.29	0.51	0.21
B32	B	2.4	2018	3.98	24	13	15.87	0.60	0.18

B4	B	1.8	2017	4.64	51	9	6.43	0.11	0.73
B5	B	1.6	2017	4.30	114	15	11.60	0.48	0.42
B6	F	16.6	2017	4.30	34	16	45.63	0.27	0.17
B7	B	1.5	2017	5.73	42	15	43.88	0.34	0.28
B8	F	14.4	2017	3.98	46	23	78.99	0.25	0.07
B9	B	2	2017	5.73	63	20	37.76	0.41	0.36
BB1	F	14.4	2019	4.64	26	13	62.48	0.43	0.25
BB2	F	12.6	2019	3.98	22	14	38.07	0.43	0.23
BB3	F	10.1	2019	3.67	19	12	51.07	0.42	0.36
BB4	F	12.2	2019	2.81	15	11	23.66	0.34	0.28
BB5	B	1.5	2019	5.73	55	14	11.48	0.81	0.10
BB6	B	2.4	2019	3.67	54	12	5.16	0.23	0.37
BB7	B	1.6	2019	4.30	60	20	27.84	0.79	0.07
BB8	B	2.1	2019	3.67	29	16	13.80	0.15	0.31
PB1	F	14.5	2020	4.99	31	16	72.82	0.50	0.42
PB10	B	2	2020	5.35	122	16	22.02	0.11	0.37
PB2	F	14.1	2020	5.73	29	14	17.28	0.50	0.36
PB3	F	12.8	2020	3.98	26	13	21.38	0.31	0.51
PB4	B	2.3	2020	3.98	60	15	6.91	0.02	0.56
PB5	F	12.2	2020	3.52	64	21	159.35	0.36	0.44
PB6	B	1.8	2020	3.37	106	13	14.25	0.09	0.72
PB7	B	2.3	2020	4.47	94	13	5.82	0.30	0.55
PB8	B	1.8	2020	3.08	100	19	30.43	0.13	0.59
PB9	F	13	2020	2.81	14	10	10.21	0.49	0.19

TABLE S2: Percent cover of benthic categories across all sampling stations in Mo’orea, French Polynesia.

Substrate type	Mean proportional cover
Bare	0.004
CCA	0.118
Dead_coral	0.021
Turf_algae	0.314
Live_coral_branching	0.115
Live_coral_encrusting	0.063
Live_coral_massive	0.162
Macroalgae	0.190
Other_inverts	0.004
Sand_rubble	0.007

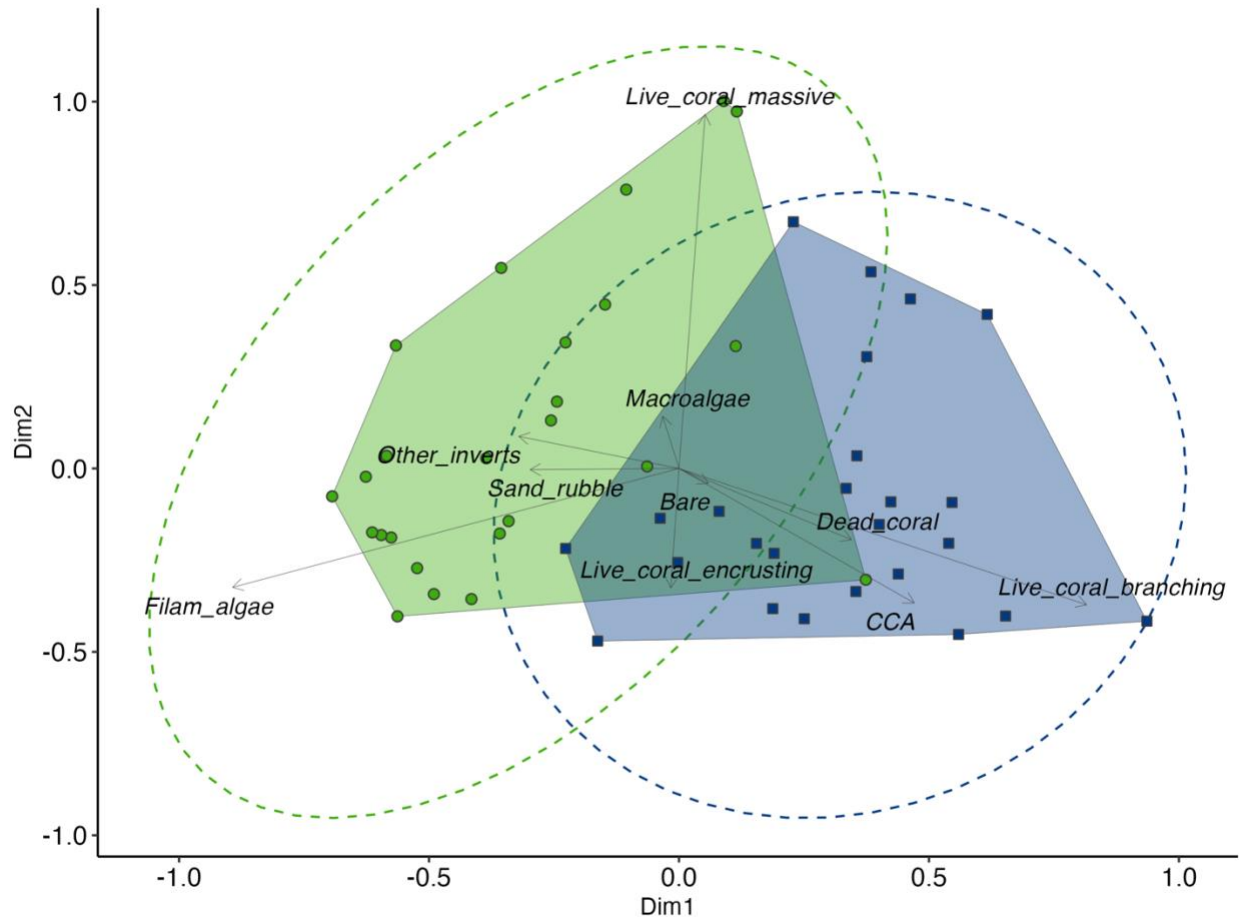


FIGURE S1: Benthic community composition in Mo'orea, French Polynesia. The Principal Coordinate Analysis (PCoA) shows the distribution of sampling stations based on their benthic cover. Convex hulls and ellipses show the spread and 95% confidence ellipses of the data in two-dimensional space, respectively (green = backreef; blue = forereef). Thin vectors and labels denote the species loadings.

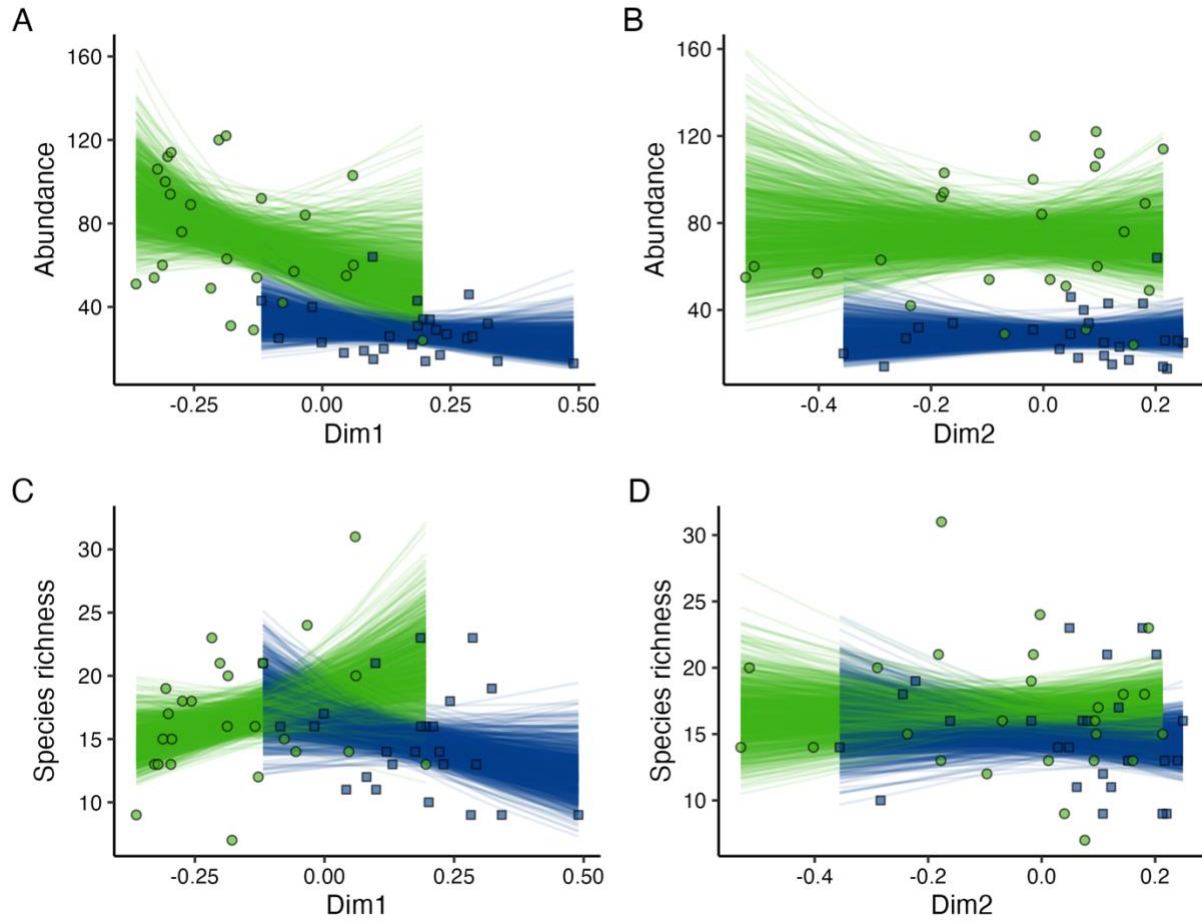


FIGURE S2: Relationships between benthic community composition (Dim1 and Dim2 from the PCoA) and cryptobenthic fish community structure (top = abundance, bottom = species richness). PCoA scores were obtained from the ordination (cf. Fig S1). Plots show the raw data (symbols) and 1,000 predicted posterior fits (lines) from Bayesian Generalized Linear Models examining the effects of Habitat (green = backreef, blue = forereef) and the two MDS axes on the two community metrics.

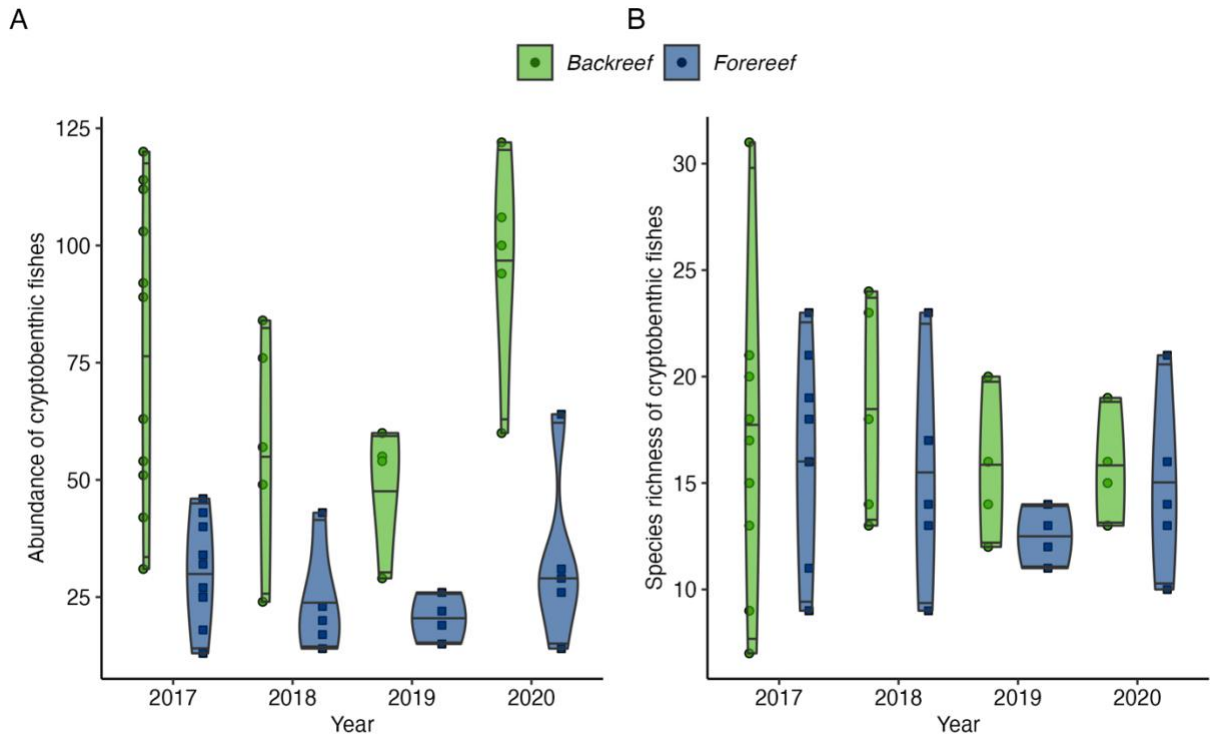


FIGURE S3: Cryptobenthic fish abundance (A) and species richness (B) across habitats (colors) and years (x-axis). Shapes represent the raw data while violins show the median and 95% confidence bands of the distributions.

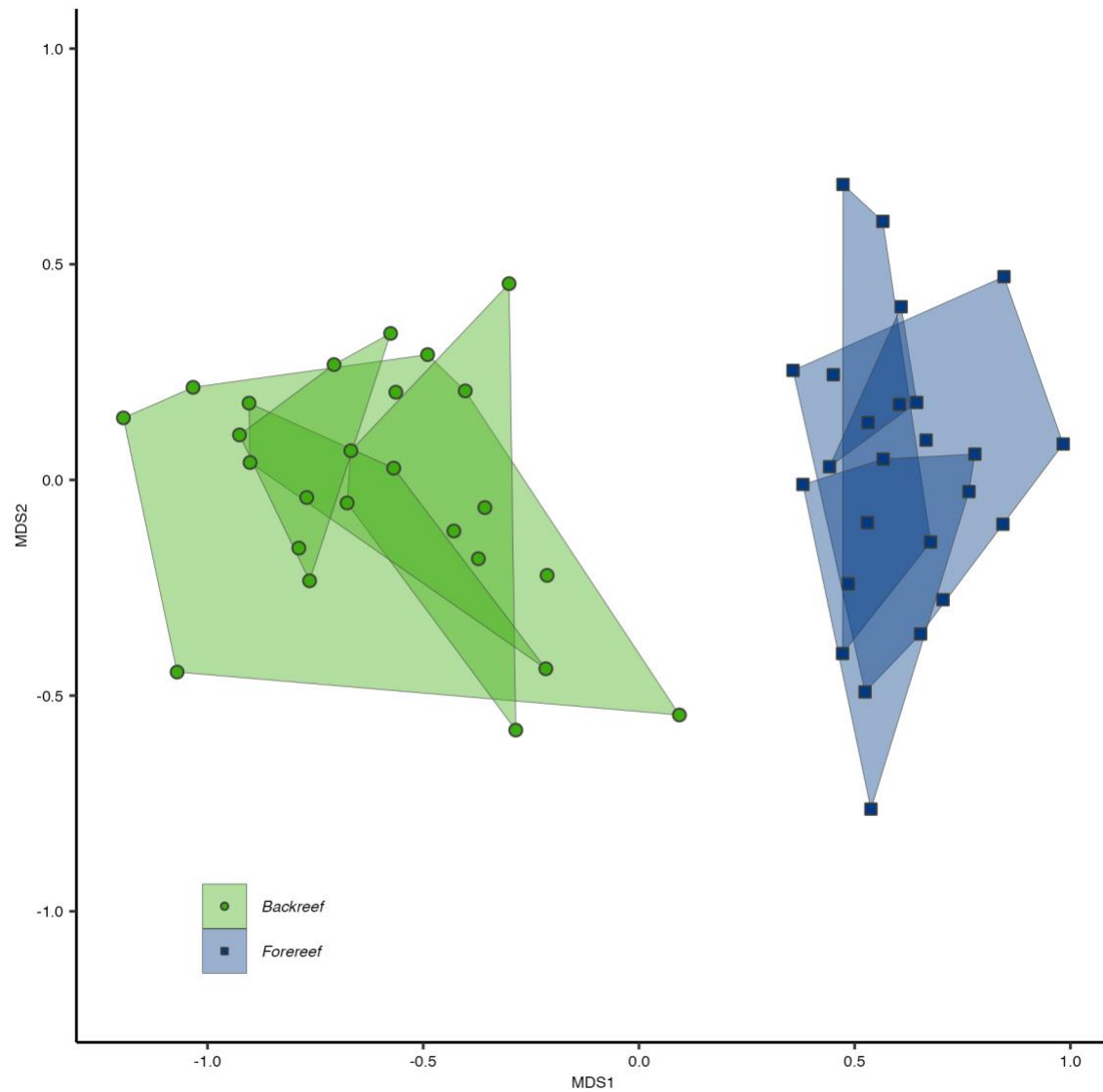


FIGURE S4: Cryptobenthic fish community composition across the two habitats and four sampling years. Separate polygons represent the four years (2017, 2018, 2019, and 2020), while colors delineate the two habitats.